DATA SCIENCE & ANALYTICS

MODULE 04 SESSION 17

# Power Query II: files that arrive as files

Append, a folder, a messy list, and Merge: one table you can trust.

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Sun 1 Nov 2026

BY THE END OF TODAY, YOU CAN

# Class objectives

01 Append queries, and say why names matter

02 Combine a folder of files into one table

03 Split columns and resolve error cells

04 Build a clean query as a Reference

05 Merge with the right join kind

EVERYONE DO

Step 1.1

# Warm-up on D16

Five questions: views, licences, Transform Data, the card that did not match, and one on UNION ALL.

INSTRUCTOR DO

Step 1.2

# How much did the shops sell in each governorate?

Two yearly files from finance, twelve files from the shops, a store list typed by hand.

shared/datasets/nakheel_stores/nakheel_store_files.zip

YOU KNOW THESE FROM SQL · TODAY IN POWER QUERY

# The same ideas, a new place

| SQL you wrote | Power Query today | Where |
|---|---|---|
| **UNION ALL** | Append queries | activity 01 |
| **many tables stacked** | a folder of files, Source.Name | activity 02 |
| **SPLIT, SAFE_CAST** | Split Column, Replace Errors | activities 04–05 |
| **a view on a table** | a Reference query | activity 04 |
| **LEFT JOIN, and … IS NULL** | Merge: Left Outer, Left Anti | activities 06–07 |

Every route today must give the shops’ completed revenue from D16: 205,338.00 dinars.

**A query** is one table and the steps that make it, listed by name in the Queries pane on the left of Power Query. Append, Reference and Merge each build a query from other queries.

PQ · 01

# Two years in one table

Finance sends store sales once a year. Stack the two files, and find out why half the revenue is missing.

| 1 · store_sales | 2 · store_files_sales | 3 · stores | 4 · store_sales_by_store |
|---|---|---|---|
| Append | Folder | Reference, Split | Merge |

DEFINITION · HOME → COMBINE → APPEND QUERIES

**Append** puts the rows of one query under the rows of another, in one table: the Power Query version of UNION ALL. Choose Append queries as new, so the two inputs stay as they are.

INSTRUCTOR DO

Step 2.1

# Append is UNION ALL, matched by name

Two yearly files, one renamed column, and a card that reads less than half.

power-query/labs/D17/01-Evr-Append-Two-Years/README.md

BEFORE → APPEND QUERIES AS NEW → AFTER · THE TWO FINANCE FILES

# Append lines columns up by name

**store_sales_2024-25.csv · 892 rows**

| order_id | store_code | order_date | status | order_total |
|---|---|---|---|---|
| 100001 | NK-KRK-01 | 2024-09-01 | completed | 22.50 |

**store_sales_2025-26.csv · 1,009 rows**

| order_id | order_date | store_code | status | amount |
|---|---|---|---|---|
| 101340 | 2025-09-01 | NK-AMM-03 | completed | 30.00 |

**indigo**: the same names, in another order · **amber**: a new name

**store_sales · 1,901 rows · 6 columns**

| order_id | store_code | order_date | status | order_total | amount |
|---|---|---|---|---|---|
| 100001 | NK-KRK-01 | 2024-09-01 | completed | 22.50 | *null* |
| 100003 | NK-AMM-01 | 2024-09-03 | completed | 47.00 | *null* |
| 101340 | NK-AMM-03 | 2025-09-01 | completed | *null* | 30.00 |
| 101341 | NK-AMM-03 | 2025-09-01 | cancelled | *null* | 25.00 |

In SQL, UNION ALL pairs by position: row 1 of 2025-26 would put **2025-09-01** under store_code.

By name: the swap did no harm; the new name became a sixth column (amber).

WHAT COMES BACK · A CARD ON SUM OF order_total · COMPLETED STORE ORDERS

# Half the revenue sits in a column the card does not add

What the shops took (D16): **205,338.00**

| **order_total, 2024-25** 97,743.00 | **amount, 2025-26** 107,595.00 |
|---|---|

| | | |
|---|---|---|
| **`Append as it came`** | the card says 97,743.00: only the first year | × |
| **`Fix it in store_sales`** | two half-empty columns, to stitch together after every append | × |
| **`Rename amount at the source`** | the card says 205,338.00, at every refresh | ✓ |

The fix goes in store_sales_2025-26, the query that reads the file: store_sales reads its two inputs again at every refresh.

EVERYONE DO

Step 2.2

# Append two years, and keep the inputs out of the model

Activity 01: append, rename amount at the source, untick Enable load on the two inputs.

power-query/labs/D17/01-Evr-Append-Two-Years/README.md

ENABLE LOAD (RIGHT-CLICK A QUERY): IS ITS RESULT LOADED INTO THE MODEL AS A TABLE OF ITS OWN?

# Rename at the source, load only the result

| Query | What you do | In the model? |
|---|---|---|
| **store_sales_2025-26** | rename amount → order_total | no: untick Enable load |
| **store_sales_2024-25** | nothing | no: untick Enable load |
| **store_sales** | nothing: it reads both again | **yes: 1,901 rows, 5 columns** |

Card, completed orders: 205,338.00, the SQL answer. Unloaded queries still run at refresh; they are not loaded as tables of their own.

THE RULE TO KEEP

# Fix the input, not the result.

A fix in the query that reads the file reaches every query built on it, at every refresh.

PQ · 02

# Twelve shop files at once

Each shop sends one file named after itself. Read the folder, not the files.

| 1 · store_sales | **2 · store_files_sales** | 3 · stores | 4 · store_sales_by_store |
|---|---|---|---|
| Append | Folder | Reference, Split | Merge |

INSTRUCTOR DO

Step 3.1

# A folder of files

Look at the file list, keep the CSV files, combine, and turn the file name into a store code.

power-query/labs/D17/02-Evr-Combine-A-Folder/README.md

# One folder, thirteen files, one table

### the folder · one row is one file

| Content | Name | Extension |
|---|---|---|
| Binary | NK-AMM-01.csv | .csv |
| Binary | NK-AMM-02.csv | .csv |
| Binary | NK-AMM-03.csv | .csv |
| Binary | NK-AQB-01.csv | .csv |
| Binary | NK-IRB-01.csv | .csv |
| Binary | NK-IRB-02.csv | .csv |
| Binary | NK-KRK-01.csv | .csv |
| Binary | NK-MDB-01.csv | .csv |
| Binary | NK-MFQ-01.csv | .csv |
| Binary | **NK-POP-01.csv** | .csv |
| Binary | NK-SLT-01.csv | .csv |
| Binary | NK-ZRQ-01.csv | .csv |
| Binary | **notes.txt** | **.txt** |



keep .csv
Combine Files

### store_files_sales · 1,901 rows

| store_code | order_id | order_date | order_total |
|---|---|---|---|
| **NK-AMM-01** | 100003 | 2024-09-03 | 47.00 |
| **NK-AMM-02** | 100006 | 2024-09-03 | 78.50 |
| **NK-AMM-03** | 100004 | 2024-09-03 | 120.00 |
| **NK-AQB-01** | 100016 | 2024-09-07 | 80.00 |

**amber**: notes.txt is not a sales file: filter it out first. **lime**: Source.Name, the file name, becomes store_code (Replace Values: .csv with nothing). **indigo**: NK-POP-01.csv, a code to remember.

Take NK-MFQ-01.csv out, Refresh: **1,855** rows. Put it back: **1,901**. Combine Files also adds Helper queries: a step added to Transform Sample file runs on every file.

With status = completed: 205,338.00, the same orders as the finance files.

EVERYONE DO

Step 3.2

# Combine the twelve shop files

Activity 02, steps 1 to 5. Browser users: your OneDrive and SharePoint folder.

power-query/labs/D17/02-Evr-Combine-A-Folder/README.md

STUDENT DO

# Which shop, which month?

Three questions a regional manager would ask, from the table you just combined.

power-query/labs/D17/03-Stu-Revenue-By-Store-And-Month/README.md

STUDENT DO

# The top shop, the best month, the odd file

| | |
|---|---|
| **INSTRUCTIONS** | store_files_sales, completed orders: 1. Which shop took the most? 2. The best and worst of the 24 months? 3. Which store code is not a shop? What does notes.txt say? |
| **HINT** | Line chart: keep Year and Month, then expand all down one level. |
| **BONUS** | A store_code slicer: do the Irbid shops peak together? |

Full steps and Check yourself: power-query/labs/D17/03-Stu-Revenue-By-Store-And-Month/README.md

REVIEW

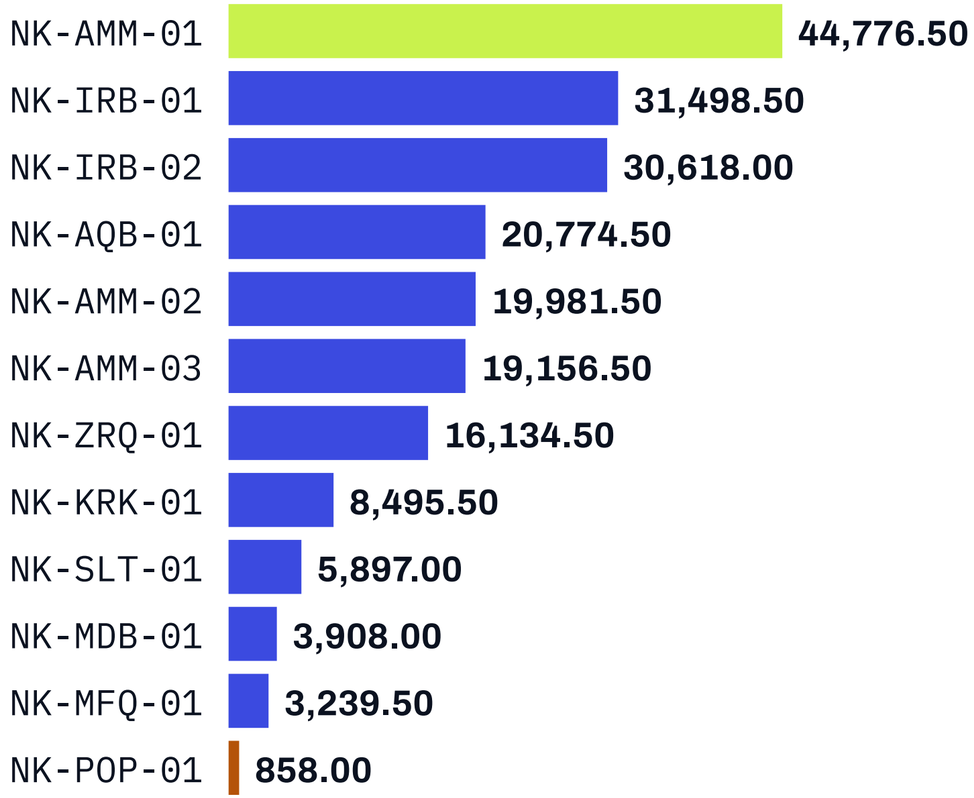
# Time's up.
# Let's review.

Which shop, which month?

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`Solved: power-query/labs/D17/03-Stu-Revenue-By-Store-And-Month/Solved/solution.md`

# Abdali (NK-AMM-01) leads; one code is a stall







The twelve bars add up to 205,338.00. NK-POP-01 (amber) is the Ramadan stall: 19 orders, 18 of them completed, 15–28 March 2025. After the break it finds no store.

THE RULE TO KEEP

# The folder is the query.

Next month’s file, or a new shop’s, needs no new work: drop it in the folder and refresh.

PQ · 03

# A store list typed by hand

Three facts in one column, sizes as text, one size unknown. Clean it in a query that reads the raw one.

| 1 · store_sales | 2 · store_files_sales | **3 · stores** | 4 · store_sales_by_store |
|---|---|---|---|
| Append | Folder | Reference, Split | Merge |

DEFINITION · RIGHT-CLICK A QUERY → REFERENCE

**A Reference** is a new query that starts from another query’s result. Change the first, and the reference follows; a Duplicate is a copy that goes its own way.

INSTRUCTOR DO

Step 4.1

# Reference, not Duplicate

Keep stores_raw exactly as the file; clean in stores, a reference to it.

power-query/labs/D17/04-Evr-Split-Reference-Errors/README.md

TWO WAYS TO START A SECOND QUERY · THEIR APPLIED STEPS

# Reference reads; Duplicate copies

**stores · right-click stores_raw → Reference**

**Source = stores_raw**

**stores_raw (2) · right-click → Duplicate**

Source = Csv.Document(…stores_raw.csv…)

Promoted Headers

Changed Type

**Head office adds a column to stores_raw.csv. Which query picks it up?**

| | | |
|---|---|---|
| **Reference** | stores reads stores_raw's result at every refresh: the new column arrives | ✓ |
| **Duplicate** | a copy of the steps, from the day it was made; it goes its own way | × |

In SQL terms: a view on a table, against CREATE TABLE … AS SELECT. Raw in, Reference out: after the break a fix in stores reaches the merge without touching it.

EVERYONE DO

Step 4.2

# Split the columns, and an error that is a fact

Activity 04: three facts into three columns, a size into a number, Jerash’s unknown size.

power-query/labs/D17/04-Evr-Split-Reference-Errors/README.md

WHAT COMES BACK · SPLIT AT EACH “ - ”

# One column, three facts, three columns

| governorate - city - area | gov_code | governorate | area |
|---|---|---|---|
| AM - Amman - Abdali | AM | Amman | Abdali |
| IR - Irbid - Irbid | IR | Irbid | Irbid |
| BA - Balqa - Salt | BA | Balqa | Salt |
| JA - Jerash - Jerash | JA | Jerash | Jerash |

The header says city; the values are a code, the governorate and the area. governorate has 9 values. Type the delimiter with its spaces, or every value keeps a space.

SPLIT size AT THE FIRST SPACE: size.1 (THE NUMBER), size.2 (THE UNIT) · size.1 AS WHOLE NUMBER

# An Error cell is a message about one value

| size (text) | size.1 as Whole Number | Then: Replace Errors → null |
|---|---|---|
| 420 m² | 420 | 420 |
| 380 sqm | 380 | 380 |
| 260 m2 | 260 | 260 |
| unknown (Jerash) | **Error** | null |

A cell-level error: the query loads, the cell says Error. A step-level error stops the query with a yellow message. Keep Errors finds error rows in any table.

QUICK CHECK

# Jerash’s size is unknown. Remove Errors or Replace Errors?

Remove Errors deletes every row with an error cell. Replace Errors puts a value of your choice, such as null, in the cell and keeps the row (SQL’s SAFE_CAST gives NULL). Which one keeps the store list true?

THE ANSWER · KEEP ERRORS → LOOK → DECIDE

# Replace the error; do not delete the shop

**Column quality on size.1 (renamed size_m2 at the end)**




Error 8% · Jerash, “unknown”

## Is Jerash still in the store list?

| | | |
|---|---|---|
| **Remove Errors** | 11 rows: Jerash is gone, and after the break no join can find it | ✗ |
| **Replace Errors → null** | 12 rows: Jerash stays, size empty; known sizes total 2,880 m² | ✓ |

EXAMPLE (lab 10, one shop’s till file): a sale typed as “93.50 JOD” is an error cell. Remove Errors drops it, and completed revenue falls from 2,032.50 to 1,939.00.

STUDENT DO

# Finish the store list

Latitude and longitude, a short name for charts, and the facilities manager’s question.

power-query/labs/D17/05-Stu-Clean-The-Store-List/README.md

**Column From Examples** makes a new column from values you type: type Abdali beside Nakheel Abdali, and Power Query writes the step that does the same for every row.

STUDENT DO

# Position, short name, floor area

| | |
|---|---|
| **INSTRUCTIONS** | In stores: 1. Split lat / long at “ / ”: latitude, longitude, Decimal Number. 2. Column From Examples on store_name: Abdali, University Street. 3. opened is a Date. 4. Square metres in Amman, and in all shops? |
| **HINT** | Errors in latitude? Change Type → Using Locale → English (United States). |
| **BONUS** | Column From Examples on the raw size: what about unknown? |

Full steps and Check yourself: power-query/labs/D17/05-Stu-Clean-The-Store-List/README.md

REVIEW

# Time's up. Let's review.

The store list

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`Solved: power-query/labs/D17/05-Stu-Clean-The-Store-List/Solved/solution.md`

THE ANSWERS · STORES, 12 ROWS

# A list you can filter, sum and map

| store_code | short_name | governorate | latitude | longitude | size_m2 |
|---|---|---|---|---|---|
| NK-AMM-01 | Abdali | Amman | 31.9632 | 35.9106 | 420 |
| NK-IRB-01 | University Street | Irbid | 32.539 | 35.856 | 310 |
| nk-irb-02 | Irbid City Centre | Irbid | 32.5556 | 35.85 | 240 |
| NK-JRS-01 | Jerash | Jerash | 32.2808 | 35.8993 | null |

Floor area: Amman 1,060 m² · all shops 2,880 m² (Jerash unknown). Look again at row 3’s code.

# Break

Stretch, then Merge

After the break: the sales meet the store list.

PQ · 04

# Sales with their store

Only the store list knows the governorate. Join it to the sales, and read the match count first.

| 1 · store_sales | 2 · store_files_sales | 3 · stores | **4 · store_sales_by_store** |
|---|---|---|---|
| Append | Folder | Reference, Split | Merge |

DEFINITION · HOME → COMBINE → MERGE QUERIES

**Merge** finds, for each row of the first query, the rows of the second with the same key, and attaches them as a table you expand: the Power Query version of a JOIN.

INSTRUCTOR DO

Step 5.1

# Merge is a JOIN; read the match count

store_sales on the left, stores on the right, by store_code, Left Outer.

power-query/labs/D17/06-Evr-Merge-Sales-And-Stores/README.md

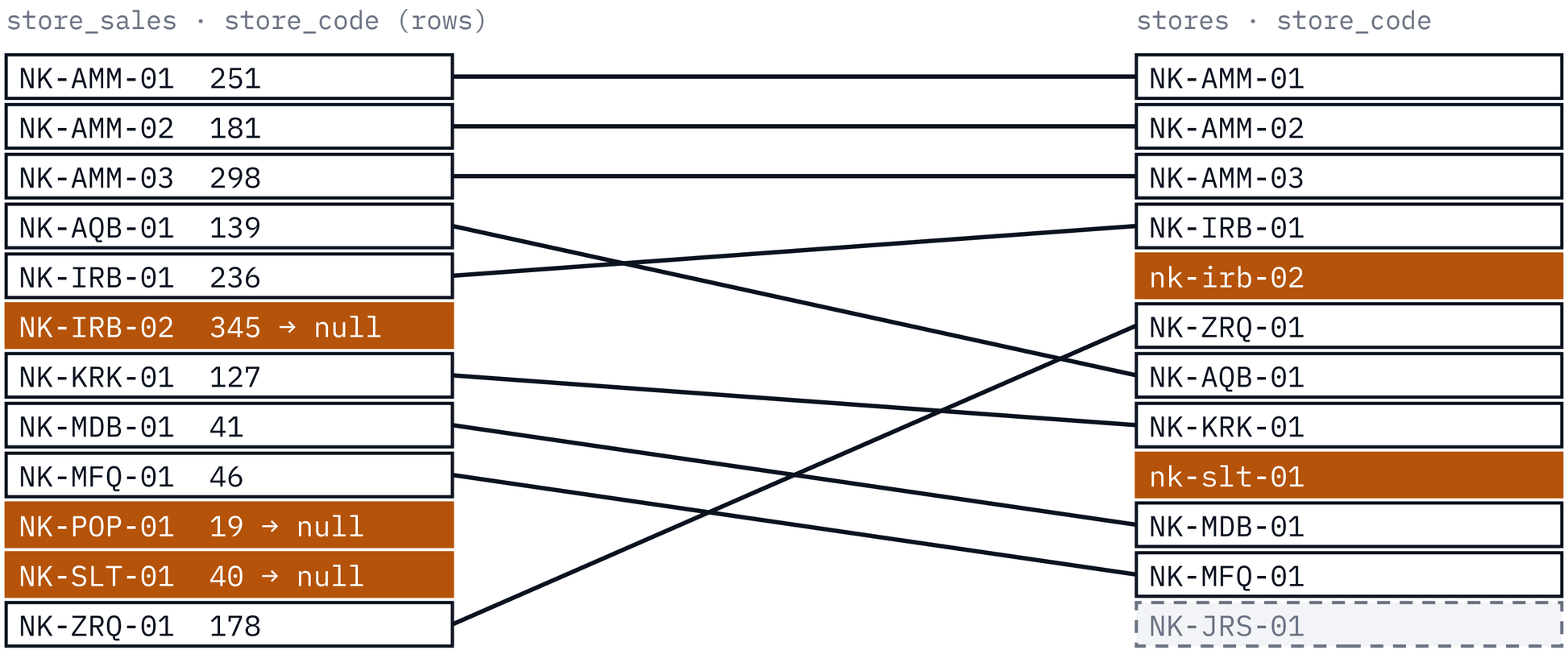
JOIN KIND IN THE MERGE DIALOG · AND THE SQL YOU WROTE

# Six join kinds, all familiar

| Join kind | Keeps | SQL |
|---|---|---|
| **Left Outer** | every left row, matches attached | LEFT JOIN |
| **Inner** | only rows that match | JOIN |
| **Full Outer** | every row of both | FULL OUTER JOIN |
| **Right Outer** | every right row | RIGHT JOIN |
| **Left Anti** | left rows with no match | LEFT JOIN … WHERE right key IS NULL |
| **Right Anti** | right rows with no match | the same, the other way |

MERGE · LEFT OUTER ON store_code · BEFORE OK, BEFORE ANY FIX

# Three codes find no store: 404 sales




**1,497**
of 1,901 match

Amber: no partner. Power Query compares text exactly: items that differ only in case are “different, not identical” (Microsoft). Dashed: Jerash, no sales yet.

AFTER OK · THE NEW COLUMN HOLDS A TABLE IN EVERY ROW

# Expand the store’s columns you need

**store_sales_by_store · the merge adds a column called stores**

| order_id | store_code | order_total | stores |
|---|---|---|---|
| 100003 | NK-AMM-01 | 47.00 | **Table** |
| 100001 | NK-KRK-01 | 22.50 | **Table** |

Each Table is that sale’s matching rows of the store list, one row when the key matches, none when it does not.



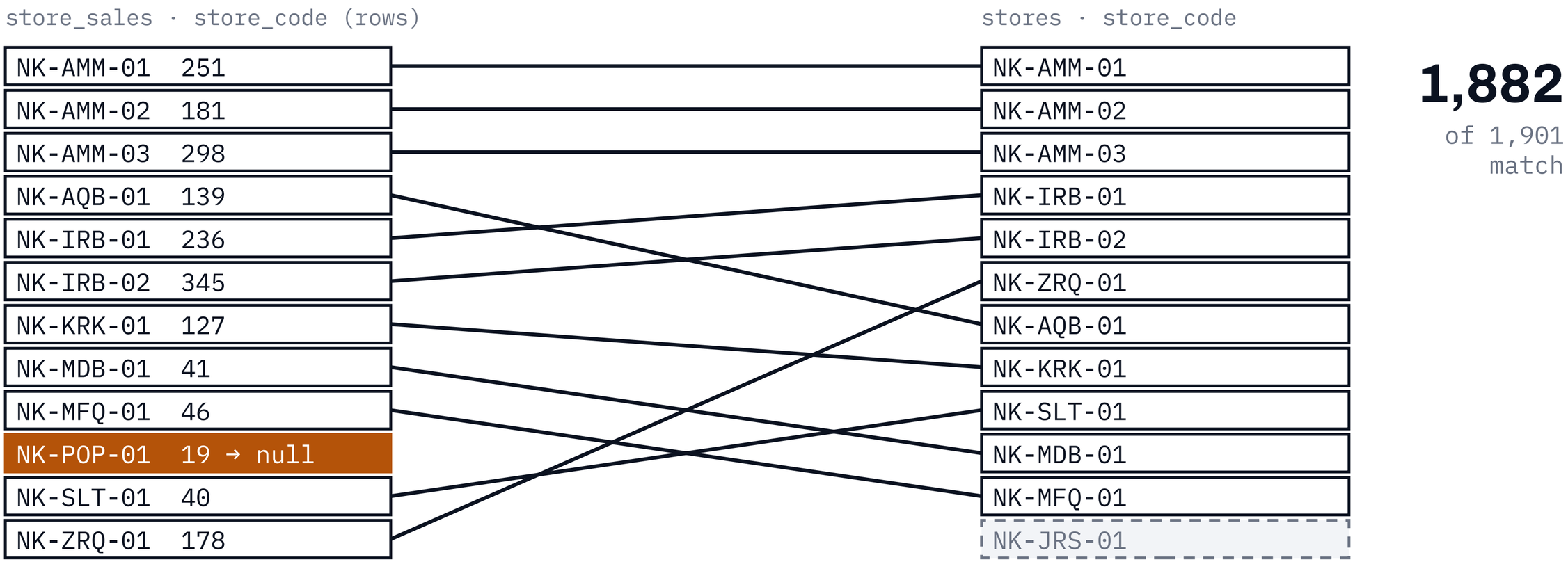
expand governorate, area, short_name · untick “Use original column name as prefix”

| order_id | store_code | order_total | governorate | area | short_name |
|---|---|---|---|---|---|
| 100003 | NK-AMM-01 | 47.00 | **Amman** | **Abdali** | **Abdali** |
| 100001 | NK-KRK-01 | 22.50 | **Karak** | **Karak** | **Karak** |

Filter governorate to (null): the 404 sales of the board before. The new columns are SQL’s right-hand columns after a LEFT JOIN.

MERGE · AFTER TRANSFORM → FORMAT → UPPERCASE ON store_code IN stores

# One sale code and one shop are left without a partner




**1,882**
of 1,901 match

The fix was made in stores, not in the merge. Left Anti: NK-POP-01, 19 sales. Right Anti: Jerash, no line. Inner: only the lines.

ILLUSTRATIVE · IF THE STORE LIST HAD NK-AMM-01 TWICE

# A repeated key on the right adds rows

Each of Abdali's 251 sales finds two store rows, so it comes back twice. **The match message does not warn you; the row count does.**

| | | |
|---|---|---|
| **Rows after the merge** | 2,152, not 1,901 | × |
| **Completed revenue** | 250,114.50, not 205,338.00: Abdali counted twice | × |
| **The check** | a key must be unique in the store list: Keep Duplicates on store_code finds it | ✓ |

D06 said the same of SQL joins: a key that repeats on one side multiplies the rows it meets.

EVERYONE DO

Step 5.2

# Merge, fix the key, chart by governorate

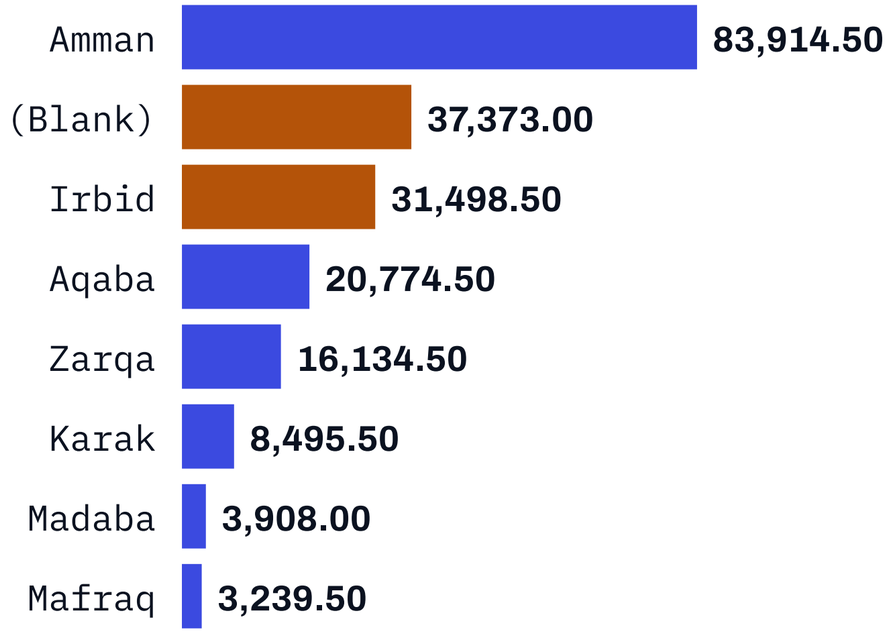
Activity 06: merge, expand, find the 404, UPPERCASE in stores, then a bar chart.

power-query/labs/D17/06-Evr-Merge-Sales-And-Stores/README.md

WHAT COMES BACK · COMPLETED REVENUE BY GOVERNORATE · LEFT OUTER

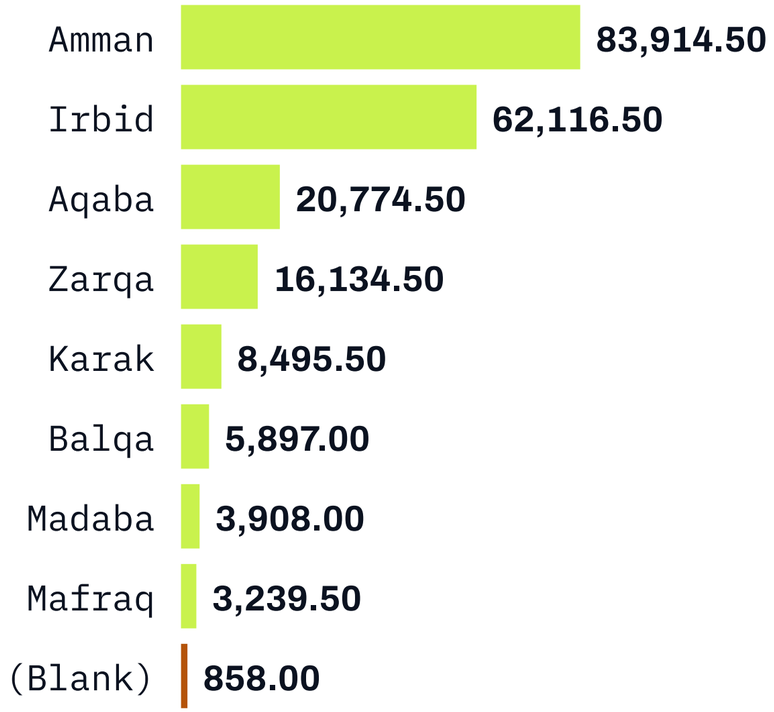
# Two lower-case codes moved 36,515.00 into (Blank)

**BEFORE · the codes as typed**




amber: wrong · Irbid lacks IRB-02 · no Balqa bar

**AFTER · UPPERCASE in stores**




Both charts add up to 205,338.00, the SQL answer from D16. The (Blank) bar left is the stall: 858.00.

STUDENT DO

# Who has no match?

An auditor’s three questions after any merge: Left Anti, Right Anti and Inner.

power-query/labs/D17/07-Stu-Sales-Without-A-Store/README.md

STUDENT DO

# Sales with no store, a store with no sales

| | |
|---|---|
| **INSTRUCTIONS** | 1. Merge Queries as New, Left Anti: which sales have no store, and their completed revenue? 2. Right Anti: which store sold nothing? 3. Inner: how many rows, what total? Which join is honest here? |
| **HINT** | Left Anti is D06’s LEFT JOIN … WHERE right key IS NULL. |
| **BONUS** | Untick Enable load on the three audit queries. |

Full steps and Check yourself: power-query/labs/D17/07-Stu-Sales-Without-A-Store/README.md

REVIEW

# Time's up.
# Let's review.

Who has no match?

---

Solved: power-query/labs/D17/07-Stu-Sales-Without-A-Store/Solved/solution.md

# Inner loses the stall; Left Outer keeps it as (Blank)

**matched to a store 204,480.00**

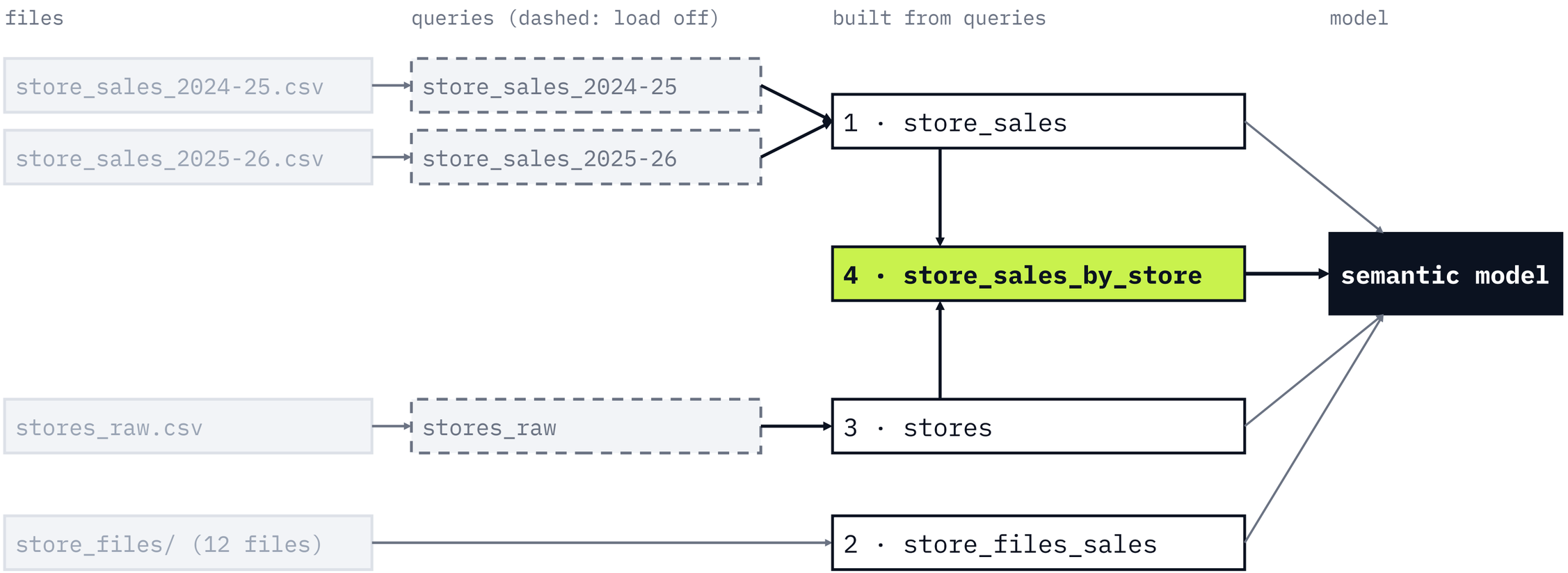
NK-POP-01, 18 completed sales: 858.00

| | | |
|---|---|---|
| **Inner** | revenue by governorate? 204,480.00: 858.00 short of the books | ✗ |
| **Left Outer** | revenue by governorate? 205,338.00, with a (Blank) bar to ask about | ✓ |
| **Left Anti** | sales with no store? 19 rows, all NK-POP-01 (18 completed) | ✓ |
| **Right Anti** | a shop with no sales? NK-JRS-01, Jerash: opens after the data ends | ✓ |

For revenue by governorate, use Left Outer and say what the blank is.

THE QUERIES YOU BUILT TODAY · NUMBERED BY PART

# Files in, one table per question out




Every arrow is a step you can name, and 205,338.00 comes out of every route. Grey arrows: queries 1 to 3 are loaded too today, to compare the routes.

THE RULE TO KEEP

# Read the match count before you click OK.

It is the cheapest check in Power Query. Then choose the join that answers the question.

STUDY LIST · IN CLASS AND IN THE LABS

# Power Query today

| In class | What it does | Activity |
|---|---|---|
| **Append queries as new** | rows of two queries in one table (UNION ALL) | 01 |
| **Enable load** | a helper query runs but is not loaded | 01, 04 |
| **Folder → Combine Files** | every file in a folder, with Source.Name | 02 |
| **Reference** | a query that starts from another query | 04 |
| **Split Column → By Delimiter** | one text column into several | 04, 05 |
| **Keep / Replace Errors** | find error cells; replace them with null | 04 |
| **Column From Examples** | a column from values you type | 05 |
| **Merge queries as new** | a join: Left Outer, Inner, the anti joins | 06, 07 |

| Lab | After class, self-guided, not submitted |
|---|---|
| 08 | **Manage Parameters: one place for a folder path; a status parameter** |
| 09 | **JSON → To Table, expand records and lists; one row per item** |
| 10 | **Remove Top Rows, Use First Row as Headers, errors that are sales** |
| 11 | **Unpivot Other Columns, Group By, a merge on two keys (optional)** |

CLOSE

# What you can do now

- Append queries; names must match
- Combine a folder of files
- Split columns; fix error cells
- Merge, and read the match count

**NEXT · D18** Queries become a model: sales and stores related, not merged; facts, dimensions, dates.